

Evaluating calibration-free two-view 3D pose fusion for exploratory repeated-cycle gymnastics analysis

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Abstract

Markerless three-dimensional motion analysis from video is attractive for sports settings, but monocular estimates remain dependent on camera view, occlusion and depth ambiguity. We evaluated how reliably two temporally aligned monocular pose streams can be reused when camera calibration and independent three-dimensional labels are unavailable. Face- and side-view poses were transformed to a pelvis-centred body frame, after which a view-order-invariant temporal network predicted a bounded correction using rotation, skeletal and temporal self-supervision. In 137 participants performing a repeated gymnastics movement, body-frame averaging reduced mean distance to a same-video triangulated pseudo-reference by 47.25% relative to world-coordinate averaging; several aligned deterministic variants performed similarly. On 14 held-out participants, the complete learned model reached 60.78 mm mean per-joint position error after similarity alignment and hip centring and was statistically indistinguishable from arithmetic averaging (60.85 mm) and the preceding learned ablation (60.80 mm). A camera-assisted comparator using estimated camera rotation reduced the held-out mean from 60.83 to 59.08 mm, although both estimates used same-video evidence. External testing on 199 Unity samples showed no zero-shot advantage over direct three-dimensional averaging (178.506 versus 166.537 mm), whereas calibrated two-dimensional triangulation reached 30.259 mm. Person-disjoint inference yielded 928 cycles from 80 elderly-labeled and 57 student-labeled participants. Cohort contrasts in angular speed and dimensionless jerk changed materially with the pose source. Coordinate canonicalization, rather than the learned correction, provided the clearest in-domain engineering gain. The resulting representation supports exploratory sequence analysis, not absolute biomechanical measurement or causal conclusions about ageing.

Keywords: markerless motion capture, self-supervised learning, data fusion, human movement, trunk rotation, repeated measures

047 **Article counts:** approximately 3,000 words in the main body; 2 figures; 5 tables.
048 Online Resource 1 contains 8 supplementary tables.

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050 1 Introduction

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052 Three-dimensional (3D) motion analysis can support technique assessment, screen-
053 ing and feedback in sport, but laboratory motion capture is not always available
054 where movements are practised. Video-based systems lower the capture burden. Mod-
055 ern monocular estimators can recover a dense articulated body from one ordinary
056 camera; SAM 3D Body, for example, predicts a full-body mesh and Momentum
057 Human Rig skeleton from an image [1]. A monocular 3D estimate nevertheless inherits
058 depth ambiguity, self-occlusion, motion blur, truncation and uncertain scale. Face and
059 side cameras observe different failures, so two predicted pose streams are potentially
060 complementary but cannot be averaged safely in their original coordinates.

061 Most multi-view methods combine images or two-dimensional (2D) evidence using
062 known or estimated camera geometry [2–5]. Other self-supervised approaches learn
063 from multi-view consistency but still begin from images or 2D poses [6–8]. These
064 methods do not directly address a common applied situation: complete monocular
065 3D sequences have already been generated, temporal correspondence is known, but
066 calibration and independent 3D labels are unavailable. Reprocessing the images with
067 a calibrated system may be impossible after field collection.

068 The engineering problem is especially relevant to rotation-dominant movement. An
069 output can be visually smooth and have low coordinate disagreement while attenuat-
070 ing the trunk rotation or peak angular speed of interest. Evaluation must therefore
071 distinguish coordinate harmonization from learned correction and determine whether
072 downstream movement findings remain stable when the pose representation changes.

073 This study evaluates three linked questions. First, how much of the in-domain
074 agreement with a same-video pseudo-reference follows from coordinate canonicaliza-
075 tion rather than learned fusion? Second, does the complete self-supervised model
076 satisfy its declared symmetry, residual and cycle-motion constraints while maintain-
077 ing held-out agreement, and how far does it transfer to independent native 3D?
078 Third, are between-cohort and within-person repeated-cycle findings stable across
079 pose sources? We contribute: (1) an applied post-estimation fusion formulation; (2) a
080 pelvis-centred, rotation-aware and view-order-invariant temporal model trained with
081 self-supervision; (3) a leakage-controlled evidence hierarchy separating same-video
082 pseudo-reference agreement from limited external native-3D testing; and (4) a down-
083 stream representation-sensitivity case study using mixed models. Because independent
084 motion capture and participant-level demographic covariates are unavailable, the study
085 does not claim absolute biomechanical accuracy, clinical validity or causal effects of
086 ageing.

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2 Methods

2.1 Study data and processing workflow

The private dataset comprised paired face- and side-view videos from 137 participants performing repeated cycles of one gymnastics movement. Dataset labels identified 80 participants as elderly and 57 as students. Exact ages, sex, health status, anthropometry and training history were not available for the present analysis; the terms *elderly-labeled* and *student-labeled* are therefore used as observational group names.

Each video was processed independently with the fixed SAM 3D Body estimator to produce $J = 70$ 3D keypoints and validity masks. An upstream procedure estimated one side-to-face temporal offset for each participant and recorded the frame bounds of each complete cycle. Fusion used this stored offset without re-estimating correspondence. A separate calibrated pipeline triangulated SAM 3D Body 2D keypoints from the same videos. Triangulated sequences were accessible only to the evaluation software and were never read during fusion training, validation, checkpoint selection or inference. They are consequently treated as a same-video *pseudo-reference*, not independent ground truth.

Ethics statement—author verification required before submission. The institutional ethics approval or exemption identifier and the confirmed participant-consent statement were not recoverable from the analysis repository. The author must replace this sentence with the approved wording before the manuscript is submitted.

2.2 Fusion method

Let $\mathbf{X}^f, \mathbf{X}^s \in \mathbb{R}^{T \times J \times 3}$ denote temporally aligned face- and side-view keypoints for a trial. Each view is canonicalized independently. The pelvis origin $\mathbf{o}_t^v$ is removed, and orthonormal axes are constructed from the hip direction and the pelvis-to-thorax direction. The robust trial scale a^v is the median valid pelvis–thorax distance. With body-frame rotation $\mathbf{R}_{p,t}^v \in \text{SO}(3)$, the canonical point is

$$\tilde{\mathbf{x}}_{t,j}^v = \frac{(\mathbf{x}_{t,j}^v - \mathbf{o}_t^v) \mathbf{R}_{p,t}^v}{a^v}. \quad (1)$$

The row-vector convention in Eq. 1 maps each camera-dependent pose into a pelvis-centred, trial-scaled body frame without estimating camera geometry. Degenerate transforms retain the last observable frame for continuity but remain masked from supervision.

A thorax frame is constructed from the shoulder direction and a thorax-to-neck direction. Relative pelvis–thorax rotation is $\mathbf{R}_{ph,t}^v = (\mathbf{R}_{p,t}^v)^\top \mathbf{R}_{h,t}^v$. Signed axial rotation and circular finite differences provide angle, angular velocity and angular acceleration. Physical timestamps, rather than an assumed constant frame interval, are used for derivatives.

Fixed frame quality combines valid-joint coverage with robust median-absolute-deviation penalties for shoulder width, hip width, torso length, bone rigidity, angular acceleration and a degenerate trunk frame. For quality q_t^v and joint-validity mask $M_{t,j}^v$,

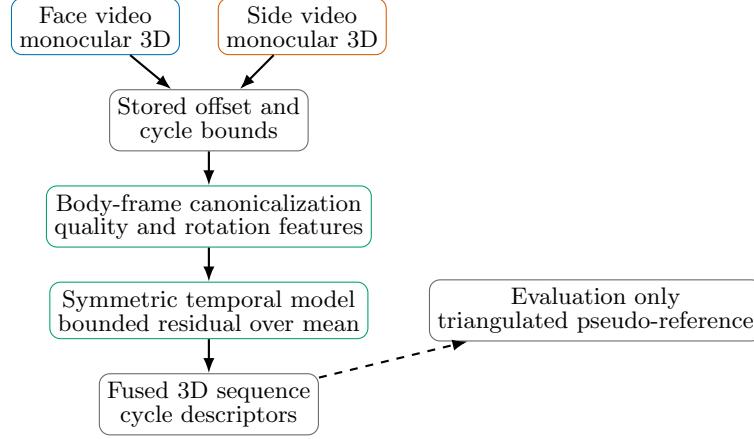


Fig. 1 Applied workflow and supervision boundary. Triangulated 3D is resolved only by the evaluation layer and cannot influence model training or selection.

the conservative base pose is

$$\mathbf{B}_{t,j} = \frac{q_t^f M_{t,j}^f \tilde{\mathbf{x}}_{t,j}^f + q_t^s M_{t,j}^s \tilde{\mathbf{x}}_{t,j}^s}{q_t^f M_{t,j}^f + q_t^s M_{t,j}^s + \epsilon}. \quad (2)$$

A point observed in only one view passes through unchanged. Equal weighting is used if both points are valid but both quality values are zero.

Shared encoders process per-view canonical coordinates, velocities, masks, quality and pelvis–thorax rotation features. View-order invariance is imposed by retaining only the encoded mean and absolute difference, together with symmetric cross-view disagreement. Six non-causal residual temporal convolutional blocks use dilations 1, 2, 4, 8, 16 and 32, following the long-context principle of temporal convolutional pose models [9, 10]. The network predicts a per-joint correction bounded to 0.05 normalized trial units:

$$\hat{\mathbf{x}}_{t,j} = \mathbf{B}_{t,j} + 0.05 \tanh(\mathbf{h}_{t,j}). \quad (3)$$

The residual is disabled unless both views are valid. The fused canonical pose is finally mapped back using the face-view origin, orientation and scale; this is an uncalibrated face reference, not a physical world coordinate system. Here, *auditable* denotes four inspectable properties: view-order invariance, a bounded residual, explicit missing-view pass-through and reported departure in cycle range and peak speed from the deterministic base. It does not imply biomechanical validity.

2.3 Self-supervision and training

Clean cross-view observations form pseudo-targets only where evidence is usable: closely agreeing points are quality-weighted, one clearly higher-quality view is selected, and single-view observations are retained when the other view is missing. Large unexplained disagreements are excluded. Seven reproducible corruptions—joint dropout,

block dropout, impulse noise, random-walk drift, upper-body rotation bias, frozen segments and integer frame shifts—are applied without overwriting the clean streams.

The complete model minimizes nine masked losses: corruption recovery, high-consensus identity, axial-angle agreement, full relative-rotation agreement, bone-length consistency, temporal bone rigidity, adaptive acceleration agreement, minimal residual magnitude and complete-cycle axial range. The complete-cycle term is evaluated on unpadded cycles, whereas other terms use 128-frame windows. The model has 128 hidden channels and was optimized with Adam for 100 epochs at learning rate 10^{-3} and batch size 32. Checkpoint selection averaged fixed-corruption recovery, bone variation, rotation consistency, identity and cycle-range scores; none used the triangulated pseudo-reference.

2.4 Evaluation hierarchy

The fixed person split contained 96 training, 27 validation and 14 held-out test participants. The 14-person set was the primary comparison population for both learned and camera-assisted methods. For every private pseudo-reference comparison, one static similarity transform was fitted separately to each cycle, followed by framewise hip centring. Errors were pooled within each participant before cross-participant summaries. Each learned comparison was paired against the complete model on the same participants using a person-bootstrap 95% confidence interval and a two-sided Wilcoxon signed-rank test; Holm correction covered the nine planned comparisons. Results use one training seed and therefore do not quantify optimization variance.

A secondary deterministic analysis over all 137 participants isolated the effect of coordinate representation by comparing world-coordinate, root-aligned, similarity-aligned and body-frame fusion variants. The held-out comparisons then tested learned residuals and camera-assisted fusion against canonical deterministic means. One legacy row whose joint weights were fitted to the evaluation pseudo-reference was retained only as a leakage diagnostic and was excluded from recommendations.

Two additional deterministic methods used an estimated side-to-face camera rotation as camera-assisted comparators. With the row-vector convention $\mathbf{X}^s = \mathbf{X}^f \mathbf{R} + \mathbf{t}$, each side pose was pelvis-centred, mapped to face axes by $\mathbf{R}$, averaged with the face pose and restored at the face pelvis; camera translation was not applied. A fixed quality-weighted version used the same rotation. Per-participant rotations were available for 129 participants, and cluster-consensus rotations were used for the other eight. Estimation used the participant’s paired videos and was therefore a transductive input operation, not a learned population parameter. Because the rotations and pseudo-reference share upstream camera observations, this comparison measures same-video agreement rather than independent 3D accuracy.

The coordinate and camera-assisted comparisons were regenerated from their compact fused sequences with this same protocol rather than reusing the earlier similarity-only summaries. Main camera-assisted and joint-level results used the same 14 held-out participants as the learned comparison. The complete 137-participant camera-assisted analysis was retained only as a secondary Online Resource result. Errors were first pooled within each participant and joint and then averaged across

participants. The main table reports the 20 predefined major MHR70 joints used by the analysis pipeline; Online Resource 1 reports all 70.

Limited external testing used 199 rendered samples from three sequences, one Unity avatar/environment and a fixed two-camera configuration. Sixteen mapped joints were evaluated against native Unity 3D after sequence-level similarity alignment. The private model was applied zero-shot. Calibrated triangulation used 2D observations and exact Unity cameras, so it is reported as a different input regime rather than a directly matched fusion baseline.

2.5 Downstream representation-sensitivity case study

Ten cohort-stratified outer folds separated people, not cycles, so every participant received an out-of-fold prediction. The resulting 928 eligible cycles comprised 539 elderly-labeled and 389 student-labeled cycles. Each cycle was direction-aligned and resampled to 101 phase points. Eight prespecified descriptors covered axial rotation range, 95th-percentile angular speed, peak-rotation phase, 95th-percentile trunk tilt, 95th-percentile wrist wrapping, cycle duration, log dimensionless angular jerk and leave-one-cycle-out whole-body repeatability.

For each descriptor, a cycle-level mixed model included cohort, normalized cycle position, their interaction and outer-fold fixed effects. Cycle position was centred at 0.5; therefore, the cohort coefficient estimates the elderly-minus-student contrast at the mid-repetition reference. Participant random intercepts and, when nonsingular, random cycle-position slopes accounted for repeated cycles. Holm correction was applied separately to the eight cohort coefficients and eight interactions. Within-person dispersion was the participant-level median absolute deviation (MAD); groups were compared with 10,000 participant-label permutations, participant-bootstrap confidence intervals and a separate eight-outcome Holm family. Pose-source sensitivity repeated the centred model using face-only, side-only and deterministic body-frame features.

3 Results

3.1 Coordinate-system effect and held-out learned comparison

Across all 137 participants, deterministic body-frame averaging had mean pseudo-reference error 65.25 mm, compared with 123.70 mm for world-coordinate averaging, a 47.25% reduction against this deliberately uncorrected baseline. Other leakage-free aligned variants ranged from 66.09 to 69.32 mm under the same evaluator. The evidence therefore supports coordinate normalization, rather than a particular weighting rule or learned residual, as the main source of in-domain improvement.

Table 1 shows that A6 was statistically indistinguishable from arithmetic averaging and A5. Its complete-cycle range and peak-angular-speed retention relative to the deterministic quality mean were both 1.000, but these ratios assess departure from the base rather than physical validity. All canonical averaging and learned outputs were view-order invariant to the reported precision. Validation-only corruption diagnostics did not establish a general robustness advantage for A6: its output moved more than

Table 1 Primary comparison on 14 held-out participants. Error is person-level mean $\pm$ standard deviation after one similarity alignment per cycle followed by framewise hip centring to the same-video triangulated pseudo-reference. Differences are method minus the complete model (A6); p values are Holm-adjusted paired Wilcoxon tests.

ID	Method	Error (mm)	Difference [95% CI]	p_{Holm}
A0	Face only	77.92 ± 12.52	$+17.14$ [13.69, 21.34]	0.0011
A1	Side only	75.84 ± 13.93	$+15.07$ [10.23, 19.48]	0.0018
A2	Arithmetic mean	60.85 ± 8.98	$+0.07$ [-0.50, 0.59]	1.0000
A3	Quality mean	61.03 ± 8.84	$+0.26$ [0.17, 0.35]	0.0011
A4	Spatial objectives	62.81 ± 8.86	$+2.04$ [1.71, 2.36]	0.0011
A5	Rotation/temporal	60.80 ± 8.96	$+0.02$ [-0.38, 0.43]	1.0000
A6	Complete model	60.78 ± 8.72	—	—

Table 2 Camera-assisted deterministic comparison on the same 14 held-out participants used in Table 1. The difference is method minus the calibration-free body-frame average, with a participant-bootstrap 95% confidence interval. p values are Holm-adjusted paired Wilcoxon tests. Each cycle uses one similarity alignment to the same-video pseudo-reference followed by framewise hip centring.

Method	MPJPE (mm)	Difference [95% CI]	p_{Holm}	Improved people
Body-frame average	60.829 ± 8.964	—	—	—
Extrinsic-R	59.081 ± 9.481	-1.747 [-2.921, -0.474]	0.0491	10/14
Extrinsic-R quality	59.595 ± 9.276	-1.233 [-2.383, -0.000]	0.0785	10/14

the deterministic quality mean under joint and temporal-block dropout. Freer post hoc residual variants increased error to 92.02–94.11 mm and are reported in the Online Resource. A6 is consequently retained as the prespecified complete test of the declared constraints, not as a demonstrated accuracy or robustness winner.

3.2 Camera-assisted and joint-level comparison

On the same 14 held-out participants as Table 1, estimated camera rotation reduced mean error by 1.747 mm (2.87%) relative to the deterministic body-frame average and was lower for 10 of 14 participants ($p_{\text{Holm}} = 0.0491$). Fixed quality weighting retained a 1.233 mm reduction and was also lower for 10 participants ($p_{\text{Holm}} = 0.0785$). Equal weighting had the lower descriptive mean and was retained as the camera-assisted comparator. The complete 137-participant analysis and its association with camera-fit quality are reported as secondary evidence in Online Resource 1. Neither analysis removes the shared-evidence limitation described above. The body-frame row in Table 2 was regenerated from the deterministic compact sequence, whereas A2 in Table 1 is the arithmetic output stored by the learned-inference pipeline. They cover the same participants, matched frames and valid points under the same evaluator; their 0.02-mm difference was retained rather than replacing either measured output.

Under the matched 14-participant evaluator, Extrinsic-R had a 20-joint macro mean of 40.35 mm, compared with 40.92 mm for A6 and 41.69 mm for the body-frame mean. It was descriptively lower than A6 at 11 of 20 joints and lower than the body-frame mean at all 20. The largest differences relative to A6 occurred at the left

Table 3 Per-joint agreement on the 14 held-out participants. Values are mean participant-level MPJPE in mm after sequence-level similarity alignment to the same-video pseudo-reference and framewise hip centring. Lower is better; bold indicates the lowest descriptive value in each row. Extrinsic-R is a camera-assisted comparator.

Joint	Face	Side	Body-frame mean	A6	Extrinsic-R
nose	65.3	38.8	40.8	40.0	39.9
left-shoulder	43.7	31.1	31.3	30.9	31.2
right-shoulder	40.0	29.0	29.7	27.9	27.0
left-elbow	47.1	47.9	38.4	39.1	36.8
right-elbow	47.8	47.8	38.2	37.9	35.4
right-wrist	89.3	83.2	66.4	65.6	64.4
left-wrist	84.8	72.9	62.0	63.7	59.6
neck	42.7	27.3	28.8	27.8	28.2
left-hip	6.4	8.7	6.1	6.3	5.3
right-hip	6.4	8.7	6.1	6.3	5.3
left-knee	27.5	40.9	27.7	28.3	27.2
right-knee	21.6	36.1	25.7	24.8	24.4
left-ankle	42.5	73.0	53.0	51.5	51.6
right-ankle	37.0	65.4	49.6	47.6	47.9
left-big-toe-tip	47.1	82.7	57.2	56.7	57.2
left-small-toe-tip	45.1	81.4	55.9	55.5	55.4
left-heel	45.5	77.5	57.2	55.3	55.6
right-big-toe-tip	39.9	73.0	52.1	50.1	50.6
right-small-toe-tip	40.7	75.0	54.0	51.8	52.2
right-heel	40.7	69.4	53.5	51.4	51.8
20-joint macro mean	43.1	53.5	41.7	40.9	40.4

Table 4 Limited external evaluation against Unity native 3D. Direct-3D and calibrated-2D rows use different input information.

Input	Method	Error (mm)	Angle MAE (°)
Uncalibrated 3D	World average	166.537	47.497
Uncalibrated 3D	A6, zero-shot	178.506	45.983
Calibrated 2D	Triangulation	30.259	15.198

wrist and both elbows; A6 or a single face view remained lower at several distal foot landmarks. The joint rows were not treated as 20 confirmatory hypothesis tests, and the macro mean is not a replacement for the all-70-joint person-level MPJPE.

3.3 External native-3D test

The external result changed the interpretation (Table 4). A6 did not improve zero-shot positional transfer relative to direct 3D averaging. Calibrated 2D triangulation was substantially more accurate, but used camera geometry and evidence unavailable to the post-estimation method.

Table 5 Sensitivity of significant A6 cohort coefficients to pose source. Values are elderly-minus-student coefficients from the same centred mixed model; p values are Holm-adjusted within each source.

Outcome	OOF A6	Face	Side	Body-frame mean
Log angular speed	0.3006	-0.0111	0.0843	0.0891
p_{Holm}	0.0114	1.0000	0.0337	0.0323
Log dimensionless jerk	0.4233	0.0452	0.0646	0.0657
p_{Holm}	0.0001	1.0000	0.2475	0.2324

3.4 Representation sensitivity in the repeated-cycle case study

All eight mixed models converged. At mid-repetition, the elderly-labeled cohort had a 0.3006 higher log angular-speed coefficient (95% confidence interval 0.1136–0.4876, $p_{\text{Holm}} = 0.0114$), corresponding to a ratio of 1.351, and a 0.4233 higher log dimensionless-jerk coefficient (0.2336–0.6130, $p_{\text{Holm}} = 0.000098$). The other six corrected cohort contrasts were not significant. No cohort-by-cycle-position interaction survived correction; there was therefore no evidence of different monotonic first-to-last repetition trends. Within-person jerk MAD was 0.0377 higher in the elderly-labeled cohort (bootstrap 95% confidence interval 0.0135–0.0735, $p_{\text{Holm}} = 0.0208$); no other within-person dispersion contrast survived.

Changing the pose source materially reduced the cohort coefficients (Table 5), especially for jerk. Separate participant-median permutation tests were also non-significant after correction for A6 angular speed and jerk. The stable finding from this case study is therefore the sensitivity of downstream inference to pose representation, not a representation-independent difference between the two labeled cohorts.

4 Discussion

The most robust engineering result is the effect of the coordinate system. Mapping both monocular streams into a pelvis-centred body frame nearly halved their disagreement with the same-video pseudo-reference compared with world-coordinate averaging. Once this mismatch was removed, deterministic and learned methods occupied a narrow range. The held-out comparison provides no evidence that the complete network improves positional accuracy over arithmetic averaging or A5. This is evidence for coordinate harmonization, not learned fusion superiority. A6 instead serves as the prespecified complete test of explicit requirements: view-order invariance, bounded correction, missing-view fallback and complete-cycle rotation constraints.

This distinction matters when an applied system is selected. If the goal is a simple fused sequence and compute or training data are limited, deterministic body-frame averaging is the better-supported default. The learned model is justified when its explicit audit trail and constrained motion representation are required, but the current results do not justify extra complexity on position error or corruption stability. Reference-relative range and angular-speed retention also cannot establish biomechanical validity, because their denominator is the deterministic base rather than independent motion capture.

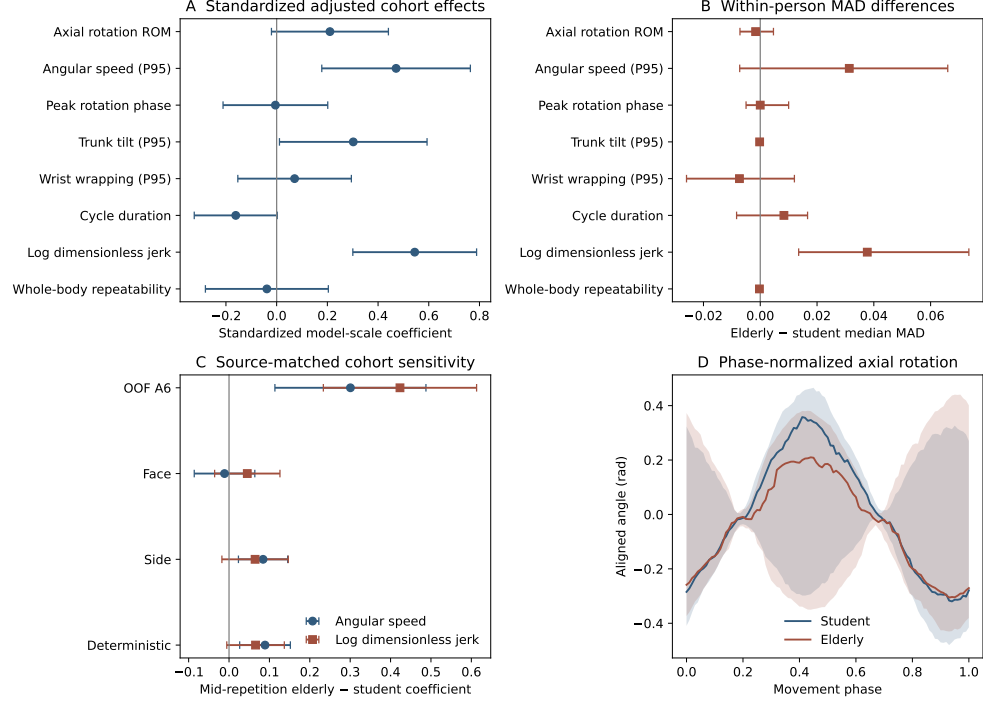


Fig. 2 Person-disjoint repeated-cycle analysis. (a) Standardized mid-repetition cohort effects. (b) Differences in participant-level within-person MAD. (c) Pose-source sensitivity for angular speed and dimensionless jerk. Error bars are 95% confidence intervals. (d) Participant-median phase-normalized axial rotation with interquartile bands.

The external-rotation comparator provides a narrower engineering result. Estimated camera rotation had lower same-video pseudo-reference error overall and at several upper-limb joints, but it requires camera estimation that the calibration-free workflow was designed to avoid. Its small advantage over A6 in the 20-joint macro mean is descriptive and does not establish a general accuracy ranking. A separate fitted-camera feature branch also failed its preregistered gate: correct-camera conditioning did not improve the frozen A6 model and was practically indistinguishable from a 30-degree wrong-camera control. Camera information was useful only in the explicit deterministic rotation operation supported by the present data.

Unity exposes a second boundary. A6 transferred worse than direct 3D averaging for position, whereas calibrated triangulation from 2D evidence was far better. Projection geometry retains information that cannot reliably be recovered after two independent monocular 3D predictions have been made. In a prospective capture where camera calibration is feasible, calibrated reconstruction should therefore be preferred. Post-estimation fusion addresses archival or low-infrastructure data where calibration is unavailable; it is not a replacement for geometry.

The repeated-cycle experiment primarily demonstrates the risk of representation-dependent downstream inference. Person-disjoint prediction avoids evaluating a

participant with a model trained on that participant, and mixed models retain cycle-level information while accounting for repeated observations. A6 produced cohort contrasts in angular speed and jerk and greater within-person jerk dispersion, but no group-specific repetition trend. The same models applied to other pose streams produced much smaller effects, and participant-median tests changed the inference. The signals may combine true movement differences, cohort composition, monocular estimation error and learned representation effects. They must not be interpreted as ageing effects, normative values, impairment markers or clinical findings.

The private pseudo-reference shares upstream 2D evidence with the candidate poses and estimated extrinsics and is scored after sequence-level similarity alignment followed by framewise hip centring. Its errors can therefore be correlated with both candidate and camera-fit errors, and the reported value is agreement with a construction rather than absolute accuracy. The primary learned and camera-assisted comparison includes only 14 participants, and the learned model uses one training seed. The Unity test contains one avatar/environment and fixed cameras. The cohort data lack individual demographic, health and training covariates. Finally, a single global time offset cannot model drift, dropped frames or rolling shutter, and the study covers one movement family and one face/side arrangement.

Future work should validate the representation against synchronized independent capture and across diverse people, cameras and movements. Public datasets such as Human3.6M, MPI-INF-3DHP and TotalCapture provide complementary settings [11–13]. For sports deployment, comparisons should include calibrated systems such as OpenCap, whose camera calibration, musculoskeletal modelling and laboratory validation support a different level of interpretation [14]. Repeated training seeds, local temporal alignment and measured participant covariates are also required before the cohort analysis can progress beyond hypothesis generation.

5 Conclusions

Calibration-free post-estimation fusion can provide a transparent sequence representation when only aligned monocular 3D pose streams remain available. Pelvis-centred canonicalization provided the clearest in-domain improvement; the complete self-supervised model matched but did not surpass strong held-out baselines, showed no general corruption-stability advantage and did not improve zero-shot external position accuracy. Estimated camera rotation gave a small camera-assisted improvement when such geometry was available, but did not alter the calibration-free model conclusion. The repeated-cycle case study further showed that between-cohort and within-person inferences changed with the pose source. The workflow is therefore best treated as an exploratory, low-infrastructure representation layer. Absolute biomechanical, causal or clinical interpretation requires calibrated or independent reference capture and richer participant information.

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Data availability	584 585
The raw videos contain identifiable participant information and are not publicly available. Code and derived anonymized keypoint data have not yet been deposited in a public repository. Release or controlled-access conditions will be determined after institutional ethics, participant-consent and privacy review. Numerical results are retained in archived experiment artifacts.	586 587 588 589 590
Ethics approval and consent to participate	591 592
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599 **Use of generative artificial intelligence**

600 During manuscript preparation, the author used OpenAI Codex for organization, lan-
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